BirdCLEF+ 2026

Status Report

Wildlife species identification from passive acoustic monitoring
in the Pantanal wetlands (Brazil). Kaggle code competition.
Metric: macro ROC-AUC. Deadline: June 3, 2026.

Current state

Best LB: 0.897 (nb19a -- Perch+MLP + top-K postproc)

Gap to 2025 winning solution (0.96): 0.063

CNN track: 5-fold ensemble trained, OOF macro-AUC 0.9792 (LB pending)

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Dataset overview

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BirdCLEF+ 2026 -- key numbers
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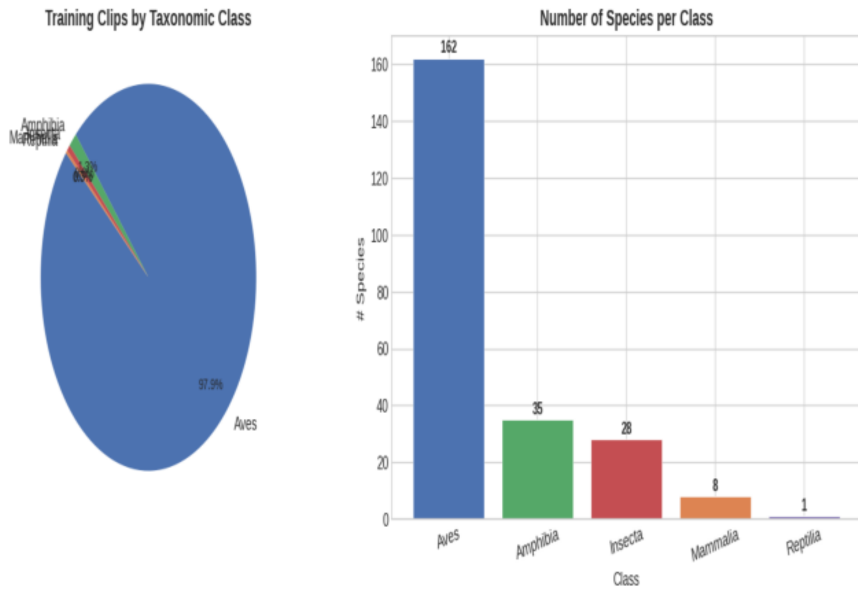
  Total training clips    : 35,549
  Species (multi-taxa)    : 234
  Aves    (birds)         : 162
  Amphibia                : 35
  Insecta                 : 28  (+25 sonotypes)
  Mammalia                : 8
  Reptilia                : 1 (Caiman, 1 clip)

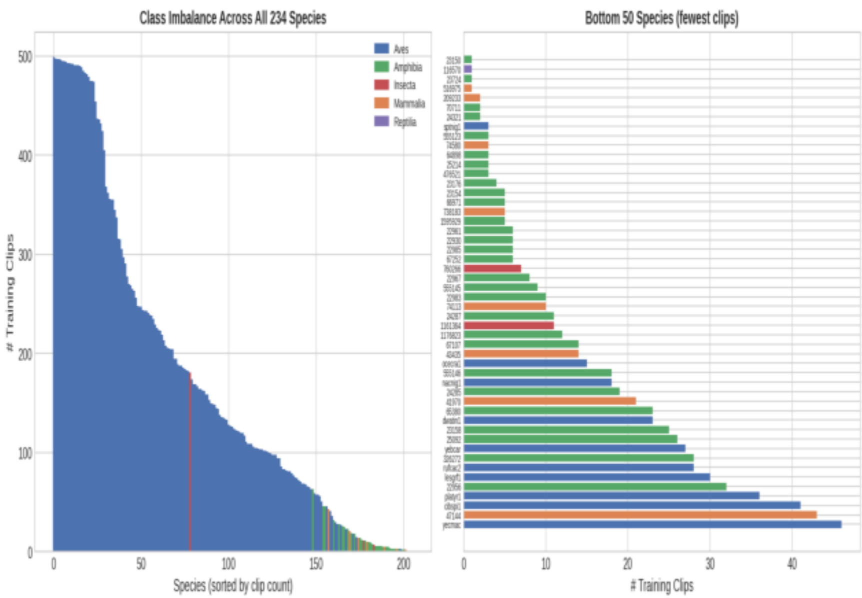
  train_audio (XC + iNat) : 35,549
  train_soundscapes       : 10,658
  Labeled segments        : 1,478 (66 files)

  Region: Pantanal, Brazil
  Lat -21.6 / -16.5,  Lon -57.6 / -55.9
  Format: OGG, 32 kHz
```

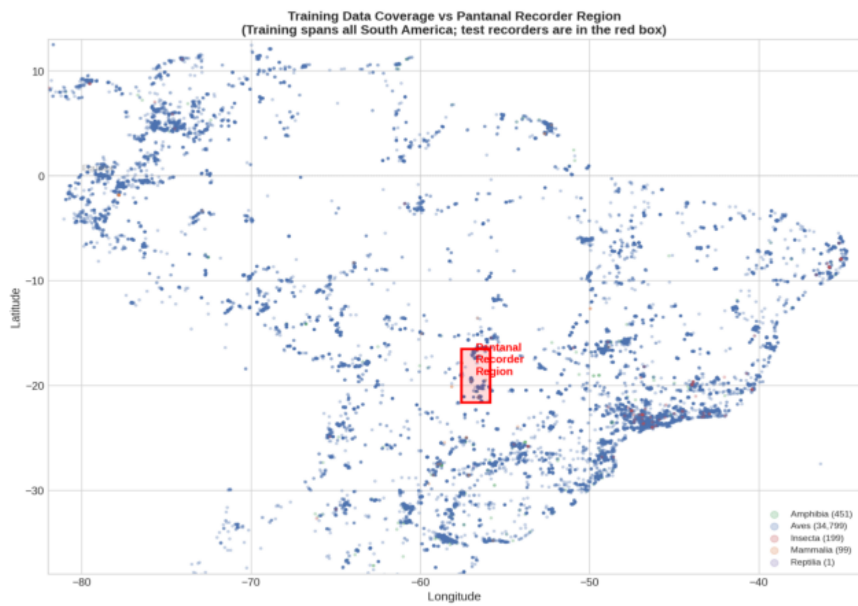
Taxa breakdown



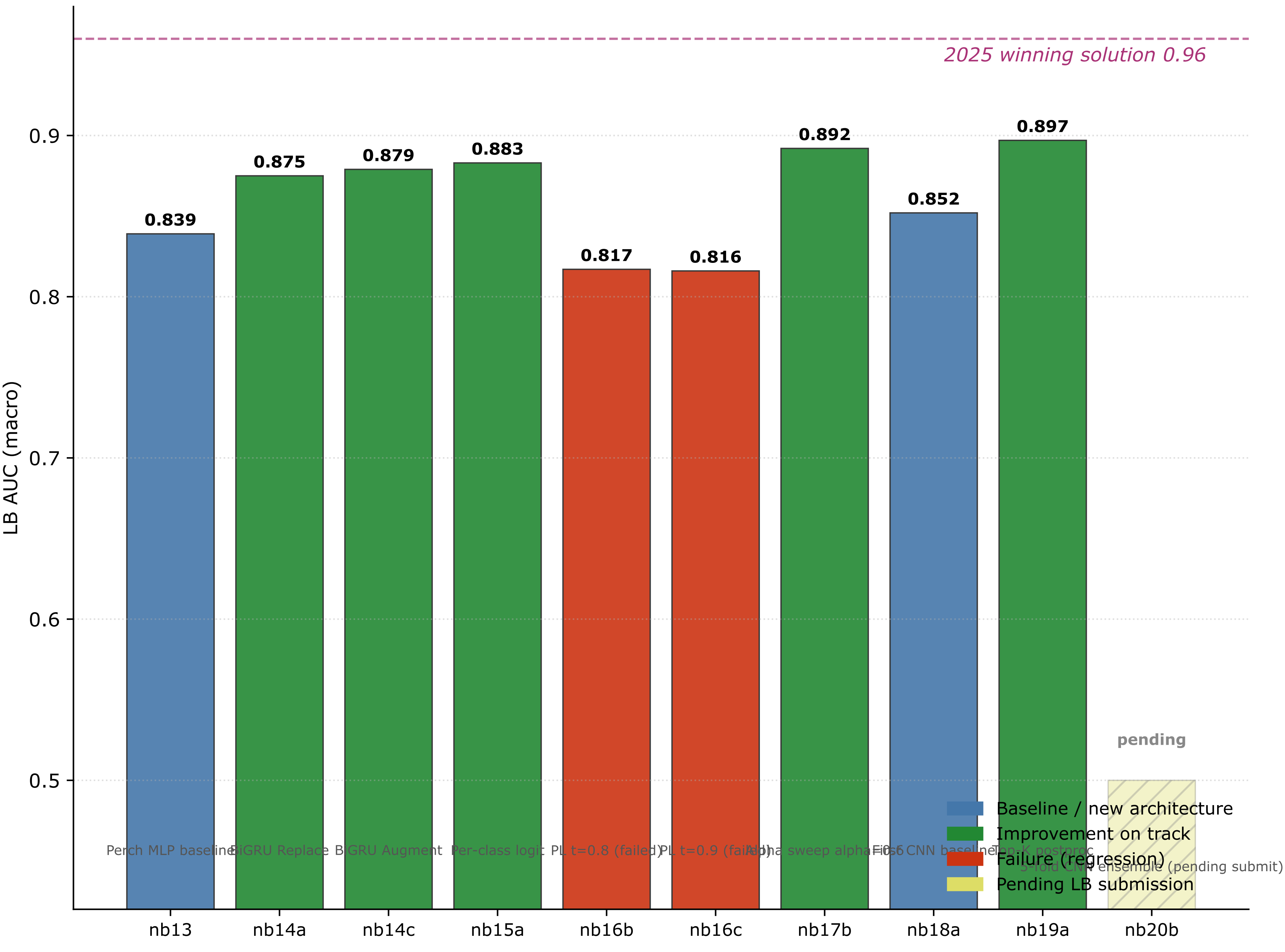
Class imbalance (long tail)



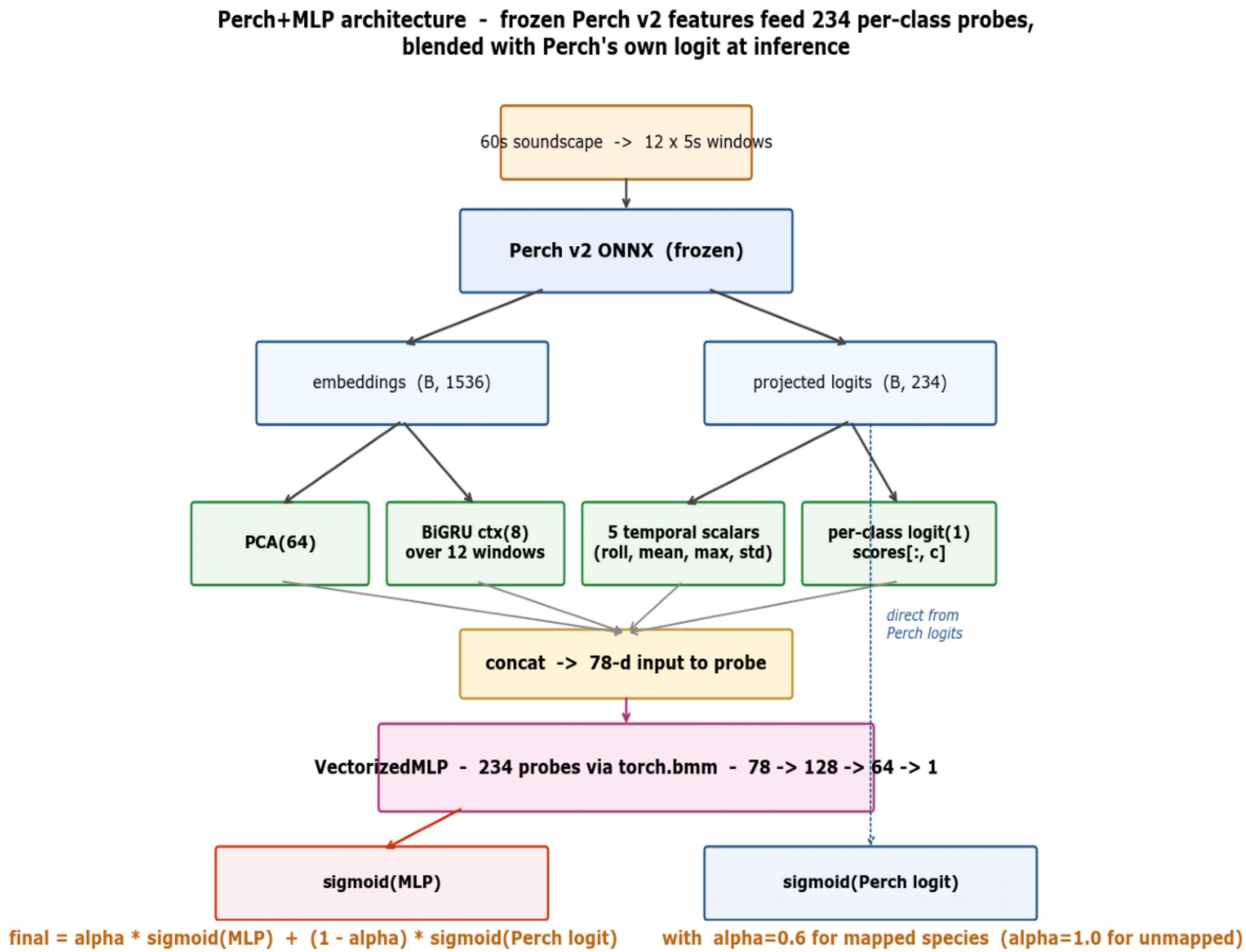
Geographic domain shift



LB progression -- 11 notebooks, two tracks

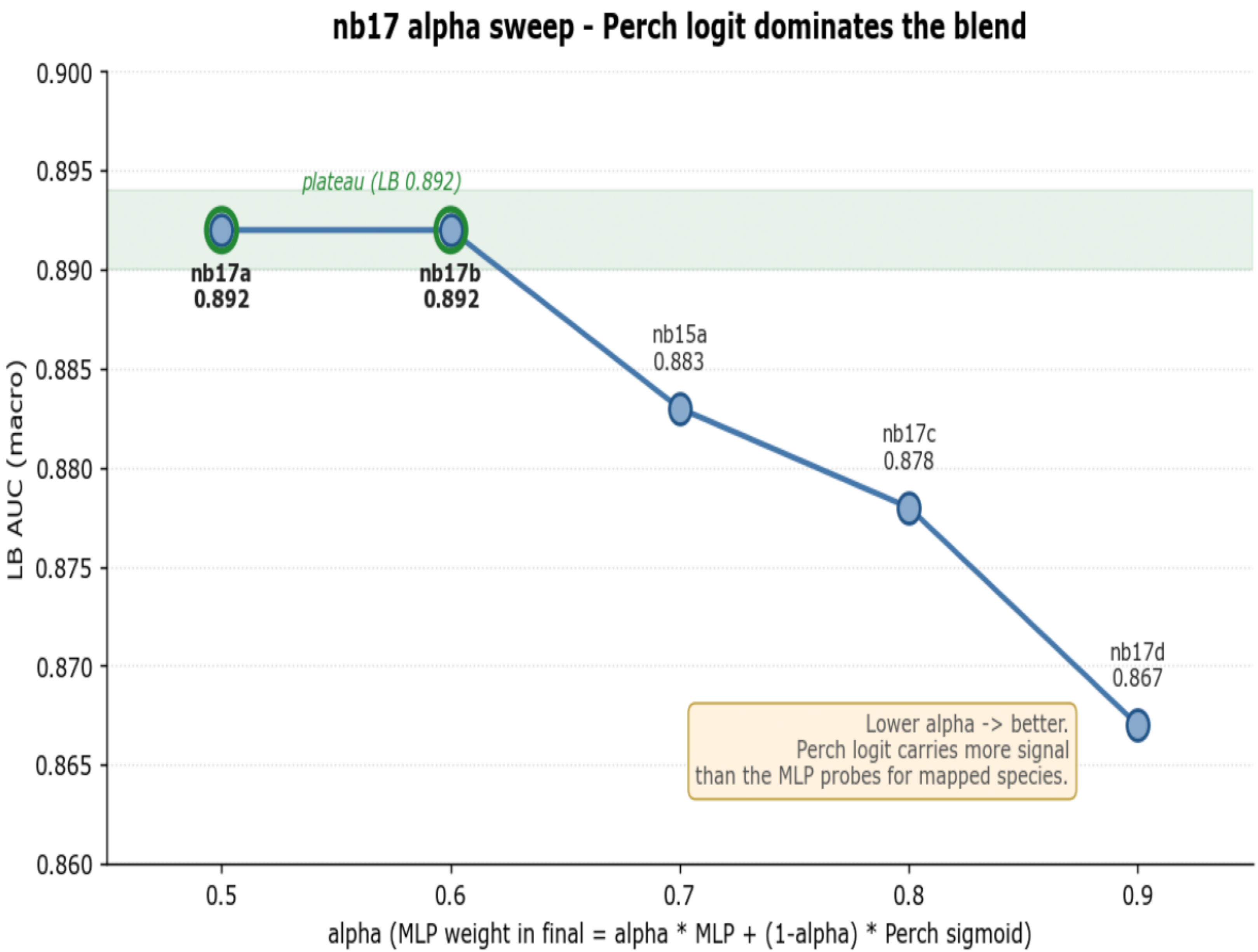


Perch+MLP architecture (LB 0.892 single, 0.897 with top-K)



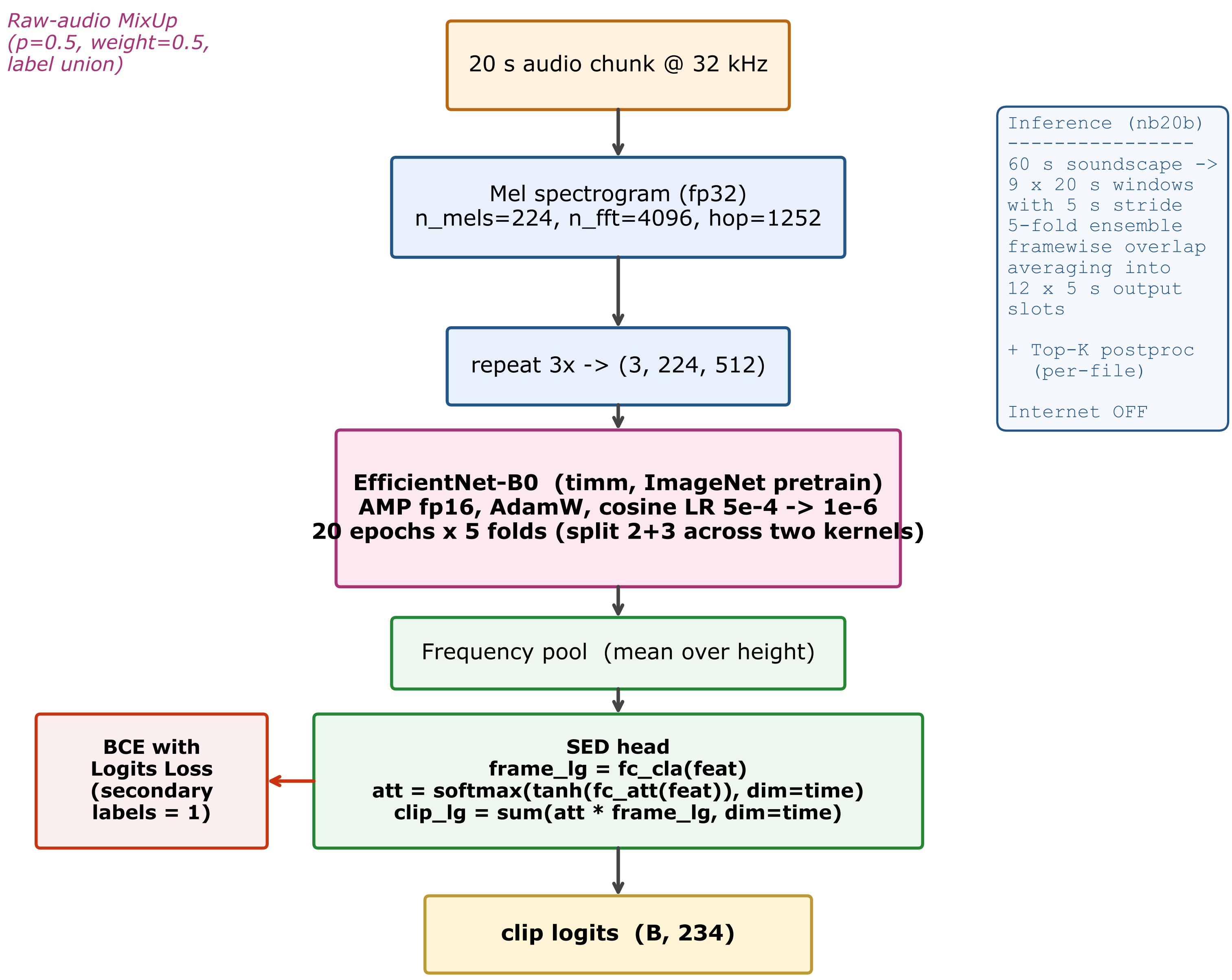
Frozen Perch v2 features feed 234 per-class probes via torch.bmm; alpha=0.6 blends the MLP correction with Perch's sigmoid at inference time.

Alpha sweep -- the architectural reframe of nb17

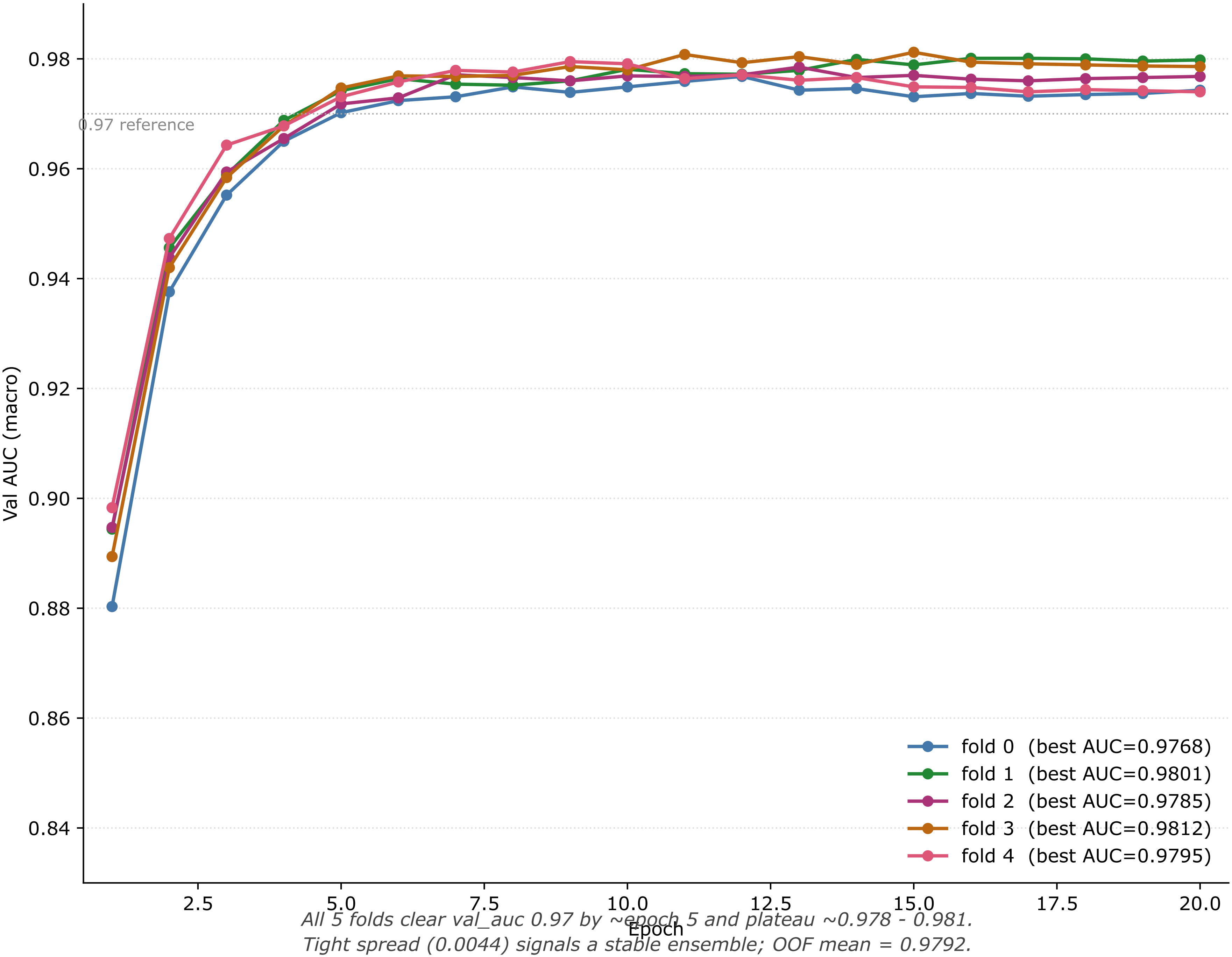


Lower alpha -> better LB. Perch's logit carries more signal than the MLP probes for mapped species, so the MLP is a correction layer on top of Perch, not an independent predictor.

CNN+SED architecture (nb20)



nb20a training curves -- 5 folds, 20 epochs each



All 5 folds clear val_auc 0.97 by ~epoch 5 and plateau ~0.978 - 0.981.
Tight spread (0.0044) signals a stable ensemble; OOF mean = 0.9792.
Caveat: StratifiedKFold-by-primary_label likely overstates the LB.

nb20a 5-fold OOF AUC (split-run training)

Fold	OOF AUC	Best epoch	Source
0	0.9768	12	run 1
1	0.9801	16	run 1
2	0.9785	-	run 2
3	0.9812	-	run 2
4	0.9795	-	run 2
Mean	**0.9792**		**5-fold**

Run 1 (alexycactus/birdclef-2026-cnn-sed-b0-run1): trained folds 0+1 in ~6.4 h on T4, saved fold checkpoints + val_preds.

Run 2 (alexycactus/birdclef-2026-cnn-sed-b0-run2): mounted run 1 via kernel_sources, trained folds 2+3+4 in ~9.7 h, ran final 5-fold ensemble inference on test_soundscapes (staging fallback).

Inference kernel (alexycactus/birdclef-2026-cnn-infer): internet OFF, 12.2 s for 5-fold ensemble + top-K postproc. Ready to submit for LB.

nb20a debug chain -- 4 failures before clean training

Version	Failure mode	Fix
v1	Slug 409 conflict	title '0.5' -> slug '0-5' != id
v2	BCELoss autocast-unsafe	swap to BCEWithLogitsLoss
v3	CUDA assert input in [0,1]	NaN slips past .clamp() -> nan_to_num
v4	Collate non-resizable	np.tile slice is a view -> np.array(copy=True)
v5 NaN	All train=nan from ep 1	fp16 FFT n_fft=4096 -> autocast(enabled=False) on mel
v5 OK	Healthy ep1 train=0.048	val_auc=0.854 in fold 0

All four failure modes added to MEMORY.md as recurring Kaggle audio-CNN gotchas to avoid in nb20c / nb21 / nb22 onwards.

Roadmap -- what's next (sorted by expected LB lever)

Notebook	What	Unblocked by	Expected LB
`nb20-infer`	Submit 5-fold CNN ensemble for LB	ready now	+0 to +0.03 vs Perch+MLP
`nb21`	Perch+MLP x CNN ensemble (rank avg or weighted)	needs nb20 LB	+0.010 to +0.025
`nb20c`	Noisy Student round 1 on the CNN	biggest single lever	+0.020 to +0.040 (1st place 2025 pattern)
`nb20c x 4`	Noisy Student rounds 2-4 (multi-iterative)	after round 1	cumulative +0.03 (1st: 0.909->0.930)
`nb20d`	Backbone diversification (EffNetV2-S, NFNet-L0)	for ensemble div	+0.005 to +0.015 in ensemble
`nb22`	Top-K post-processing variants	fast iteration	+0.005 (already applied in nb20-infer)
`nb25 (deferred)`	Xeno-Canto pretraining	high setup cost	+0.025 from 2nd place 2025
`nb27`	Specialised Amphibia/Insecta model	small but quick	+0.002 to +0.003

Full backlog: docs/plans/2026-05-14-post-nb20-experiments-backlog.md